Recursive domain equations

Lean 4 Formalization Companion – Chapter 5

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Section 5.1

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Definition 5.1.1. Let F be a functor on a category $\mathbf{C}$. An F-algebra is an object A together with a map $f : F(A) \to A$. A homomorphism from $f : F(A) \to A$ to $f' : F(A') \to A'$ is a map $g : A \to A'$ such that

$$\begin{array}{ccc} F(A) & \xrightarrow{F(g)} & F(A') \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ A & \xrightarrow[g]{} & A' \end{array}$$

commutes.

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Lemma 5.1.2. If $i : F(A) \to A$ is an initial F-algebra, then i is an isomorphism. *Proof sketch.* Let f be the unique homomorphism from $i : F(A) \to A$ to $F(i) : F^2(A) \to F(A)$. The proof uses the composition of homomorphisms displayed below:

$$\begin{array}{ccccc} F(A) & \xrightarrow{F(f)} & F^2(A) & \xrightarrow{F(i)} & F(A) \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle F(i)} & & \downarrow{\scriptstyle i} \\ A & \xrightarrow[f]{} & F(A) & \xrightarrow[i]{} & A \end{array}$$

Initiality gives $i \circ f = \mathrm{id}_A$, and the remaining equality gives the inverse law on $F(A)$.

Section 5.2

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Definition 5.2.1. A functor $F : \mathbf{D}^e \to \mathbf{D}^e$ is continuous if, for every expanding sequence $\langle (D_n)_{n\in\mathbb{N}}, (e_{mn} : D_n \to D_m)_{n\le m}\rangle$ with colimit $\langle D, (e_n)_{n\in\mathbb{N}}\rangle$, the cocone

$$\langle F(D), (F(e_n))_{n\in\mathbb{N}}\rangle$$

is a colimit of the sequence $\langle (F(D_n))_{n\in\mathbb{N}}, (F(e_{mn}))_{n\le m}\rangle$.

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Lemma 5.2.2. Let F be a continuous functor on a category $\mathbf{D}^e$ of domains. For an embedding $e : A \to F(A)$, let $\langle D, (e_n)_{n\in\mathbb{N}}\rangle$ be the colimit of

$$A \xrightarrow{e} F(A) \xrightarrow{F(e)} F(F(A)) \xrightarrow{F(F(e))} \cdots .$$

Then $D \cong F(D)$. The isomorphism is given by

$$\text{fold} = \bigvee_{n\in\mathbb{N}} e_{n+1} \circ F(e_n)^* : F(D) \to D,$$
$$\text{unfold} = \bigvee_{n\in\mathbb{N}} F(e_n) \circ e_{n+1}^* : D \to F(D),$$

and for every n,

$$F(e_n) = \text{unfold} \circ e_{n+1}, \qquad F(e_n)^* = e_{n+1}^* \circ \text{fold}\,.$$

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Definition 5.2.3. A functor $F : \mathbf{D} \to \mathbf{E}$ between categories of domains is called locally continuous if, for every pair of objects D, D', the map

$$\text{Hom}(D, D') \longrightarrow \text{Hom}(F(D), F(D')), \qquad f \longmapsto F(f),$$

is Scott-continuous.

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Proposition 5.2.4. A locally continuous functor $F : \mathbf{D} \to \mathbf{E}$ restricts to a continuous functor $F : \mathbf{D}^e \to \mathbf{E}^e$ on the corresponding categories of embeddings.

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Definition 5.2.5. A functor $F : \mathbf{D}^{op} \times \mathbf{D}' \to \mathbf{E}$, contravariant in the first coordinate and covariant in the second, is locally continuous if, for directed sets $A \subseteq \text{Hom}(D_2, D_1)$ and $A' \subseteq \text{Hom}(D_1', D_2')$,

$$F(\bigvee\nolimits^{\uparrow} A, \bigvee\nolimits^{\uparrow} A') = \bigvee\nolimits^{\uparrow}_{f\in A,\, f'\in A'} F(f, f')$$

in $\text{Hom}(F(D_1, D_1'), F(D_2, D_2'))$.

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Proposition 5.2.6. If $F : \mathbf{D}^{op} \times \mathbf{D}' \to \mathbf{E}$ is mixed-variant and locally continuous, it defines a continuous covariant functor $\widehat{F} : \mathbf{D}^e \times \mathbf{D}'^e \to \mathbf{E}^e$ by

$$\widehat{F}(D, D') = F(D, D'), \qquad \widehat{F}(e, e') = F(e^*, e')$$

for embeddings. The upper adjoint of $\widehat{F}(e, e')$ is $F(e, e'^*)$. If (e, e^*) and (e', e'^*) are embedding-projection pairs, then

$$F(e, e'^*) \circ \widehat{F}(e, e') = F(e, e'^*) \circ F(e^*, e') = F(e^* \circ e, e'^* \circ e') = \text{id},$$
$$\widehat{F}(e, e') \circ F(e, e'^*) = F(e^*, e') \circ F(e, e'^*) = F(e \circ e^*, e' \circ e'^*) \sqsubseteq \text{id}\,.$$

Thus $\widehat{F}$ maps embeddings to embeddings. Continuity follows by applying local continuity to expanding sequences in the two coordinates.

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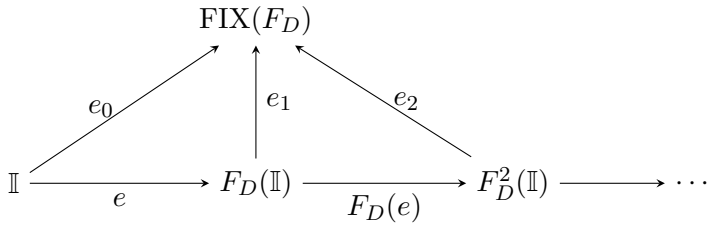
Proposition 5.2.7. Let $F : \mathbf{D}_{\perp!} \times \mathbf{E}_{\perp!} \to \mathbf{E}_{\perp!}$ be locally continuous. Then the assignment Lean

$$D \mapsto \mathrm{FIX}(F_D), \qquad F_D(E) = F(D, E),$$

extends to a locally continuous functor $\mathbf{D}_{\perp!} \to \mathbf{E}_{\perp!}$. On a strict map $f : D \to D'$, the induced map is

$$f \mapsto \bigvee_{n \in \mathbb{N}} e'_n \circ f_n \circ e^*_n,$$

where $f_0 = \mathrm{id}_{\mathbb{I}}$ and $f_{n+1} = F(f, f_n)$. The canonical solution of $X \cong F(D, X)$ is obtained as the bilimit of



and similarly for D'.

Section 5.3

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Lemma 5.3.1. Let F be a locally continuous functor on a category of domains $\mathbf{D}$, and let $i : F(A) \to A$ be an isomorphism. For every F-algebra $f : F(C) \to C$ there is a least homomorphism $h_{C,A} : A \to C$. It is the least fixed point of the functional $\phi_{C,A}$ on $[A \to C]$ defined by

$$\phi_{C,A}(g) = f \circ F(g) \circ i^{-1}.$$

Least homomorphisms compose: if $j : F(B) \to B$ is also an isomorphism, then

$$h_{C,A} = h_{C,B} \circ h_{B,A}.$$

Proof sketch. The functional $\phi_{C,A}$ is continuous because F is locally continuous and composition is continuous. Hence its least fixed point exists as the supremum of the chain starting from $c_\perp$; induction shows that this point is below every homomorphism and is itself a homomorphism. Composition follows from Lemma 2.1.21 applied to the square whose horizontal map sends g to $g \circ h_{B,A}$, giving $\mathrm{fix}(\phi_{C,A}) = \mathrm{fix}(\phi_{C,B}) \circ h_{B,A}$.

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Theorem 5.3.2. *(Invariance, Part 1)* Let F be a locally continuous functor on a category of domains $\mathbf{D}$, and let $i : F(A) \to A$ be an isomorphism. The following are equivalent:

1. A is isomorphic to the canonical fixpoint $\mathrm{FIX}(F)$;
2. id_A is the least endomorphism of A;
3. $\mathrm{id}_A = \mathrm{fix}(\phi_{A,A})$, where $\phi_{A,A} : [A \to A] \to [A \to A]$ is defined by $\phi_{A,A}(g) = i \circ F(g) \circ i^{-1}$;
4. id_A is the only strict endomorphism of A.

Proof sketch. For $D = \mathrm{FIX}(F)$, the least endomorphism is the least fixed point of $g \mapsto \mathrm{fold} \circ F(g) \circ \mathrm{unfold}$. With the standard embeddings $e_n : F^n(\mathbb{I}) \to D$, induction gives $c_\perp = e_0 \circ e_0^*$ and $\phi^n(c_\perp) = e_n \circ e_n^*$, whose supremum is the identity by Lemma 3.3.8. The equivalence of (2) and (3) is Lemma 5.3.1. For (3) implies (4), Lemma 2.1.21 is applied to the square with $H(g) = h \circ g$ for a strict endomorphism h, yielding $\mathrm{id}_A = H(\mathrm{id}_A) = h$. Finally, the homomorphisms between A and $\mathrm{FIX}(F)$ compose to the least endomorphisms, hence to identities under (4).

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Theorem 5.3.3. *(Minimality, Part 1)* The canonical fixpoint of a locally continuous functor is a sub-domain of every other fixpoint.

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Theorem 5.3.4. *(Initiality)* Let $F : \mathbf{D}_{\perp !} \to \mathbf{D}_{\perp !}$ be a locally continuous functor on a category of domains with strict functions. Then $\mathrm{fold} : F(D) \to D$ is an initial F-algebra, where D is the canonical solution to $X \cong F(X)$. *Proof sketch.* For a strict algebra $f : F(A) \to A$, Lemma 5.3.1 gives a homomorphism $h : D \to A$, and inspection of the construction shows that h is strict. Uniqueness is proved as in Theorem 5.3.2, using Lemma 2.1.21 on the commutative

square

$$\begin{array}{ccc} [D \to D] & \xrightarrow{\quad H \quad} & [D \to A] \\ {\scriptstyle \phi_{D,D}}\downarrow & & \downarrow{\scriptstyle \phi_{A,D}} \\ [D \to D] & \xrightarrow[\quad H \quad]{} & [D \to A] \end{array}$$

where $H(g) = h \circ g$.

Theorem 5.3.5. *(Finality)* Let $F : \mathbf{D} \to \mathbf{D}$ be a locally continuous functor with canonical fixpoint $D = \mathrm{FIX}(F)$. Then $\mathrm{unfold} : D \to F(D)$ is a final co-algebra. Lean

Lemma 5.3.6. Let $F : \mathbf{D}^{op} \times \mathbf{D} \to \mathbf{D}$ be a mixed-variant, locally continuous functor, and let $i : F(A, A) \to A$ and $j : F(B, B) \to B$ be isomorphisms. Then there exists a least pair of functions $h : A \to B$ and $k : B \to A$ such that the two squares Lean

$$\begin{array}{ccc} F(A,A) & \xrightarrow{F(k,h)} & F(B,B) \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle j} \\ A & \xrightarrow{h} & B \end{array} \qquad \begin{array}{ccc} F(B,B) & \xrightarrow{F(h,k)} & F(A,A) \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle i} \\ B & \xrightarrow{k} & A \end{array}$$

commute.

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Theorem 5.3.7. *(Invariance, Part 2)* Let $F : \mathbf{D}^{op} \times \mathbf{D} \to \mathbf{D}$ be a mixed-variant and locally continuous functor, and let $i : F(A, A) \to A$ be an isomorphism. The following are equivalent: Lean

1. A is isomorphic to the canonical fixpoint $\mathrm{FIX}(F)$;
2. id_A is the least mixed endomorphism of A;
3. $\mathrm{id}_A = \mathrm{fix}(\phi_{A,A})$, where $\phi_{A,A} : [A \to A] \to [A \to A]$ is defined by $\phi_{A,A}(g) = i \circ F(g, g) \circ i^{-1}$;
4. id_A is the only strict mixed endomorphism of A.

Proof sketch. The proof follows Theorem 5.3.2, with the mixed-variance step made explicit. If $D = \mathrm{FIX}(F)$ and $e_n : F^n(\mathbb{I}) \to D$ are the colimiting maps, then the cocone into $F(D, D)$ gives

$$F(e_n^*, e_n) = \mathrm{unfold} \circ e_{n+1}, \qquad F(e_n^*, e_n)^* = F(e_n, e_n^*) = e_{n+1}^* \circ \mathrm{fold}\,.$$

For the least mixed endomorphism, induction gives

$$\begin{aligned} \phi^{n+1}(c_\perp) &= \phi(\phi^n(c_\perp)) = \phi(e_n \circ e_n^*) \\ &= \mathrm{fold} \circ F(e_n \circ e_n^*, e_n \circ e_n^*) \circ \mathrm{unfold} \\ &= \mathrm{fold} \circ F(e_n^*, e_n) \circ F(e_n, e_n^*) \circ \mathrm{unfold} = e_{n+1} \circ e_{n+1}^*. \end{aligned}$$

Contravariance in the first argument is exactly what swaps e_n and e_n^*. For (3) implies (4), use Lemma 2.1.21 with $H(g) = h \circ g \circ h$.

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Theorem 5.3.8. *(Minimality, Part 2)* The canonical fixpoint of a mixed-variant and locally continuous functor is a sub-domain of every other fixpoint. Lean

Theorem 5.3.9. *(Free mixed-variant algebra)* Let $F : \mathbf{D}_{\perp!}^{op} \times \mathbf{D}_{\perp!} \to \mathbf{D}_{\perp!}$ be a mixed-variant, locally continuous functor, and let D be the canonical solution to $X \cong F(X, X)$. For every pair of strict continuous functions

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$$f : A \to F(B, A), \qquad g : F(A, B) \to B,$$

there are unique strict functions $h : A \to D$ and $k : D \to B$ such that the two squares

$$\begin{array}{ccc} F(B,A) & \xrightarrow{F(k,h)} & F(D,D) \\ {\scriptstyle f\uparrow} & & {\scriptstyle \uparrow\mathrm{unfold}} \\ A & \xrightarrow{h} & D \end{array} \qquad\qquad \begin{array}{ccc} F(D,D) & \xrightarrow{F(h,k)} & F(A,B) \\ {\scriptstyle \mathrm{fold}\downarrow} & & {\scriptstyle \downarrow g} \\ D & \xrightarrow{k} & B \end{array}$$

commute.

Section 5.4

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Definition 5.4.1. A relation $R \subseteq D^n$ on a pointed domain D is called admissible if it contains the constantly-bottom tuple and is closed under suprema of ω-chains. Write $\mathcal{R}^n(D)$ for the set of all admissible n-ary relations on D, ordered by inclusion; unary relations of this kind are admissible predicates. If $f : D \to E$ is strict and continuous, then it acts pointwise on relations by

$$f^{rel}(R) = \{\langle f(x_1), \ldots, f(x_n)\rangle \mid \langle x_1, \ldots, x_n\rangle \in R\}.$$

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Proposition 5.4.2. For dcpo's D and E, and admissible n-ary relations R on D and S on E, the set

$$\{f \mid f^{rel}(R) \subseteq S\}$$

is an admissible predicate on the strict function space $[D \to E]$.

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Definition 5.4.3. Let $F : \mathbf{D}_{\perp!}^{op} \times \mathbf{D}_{\perp!} \to \mathbf{D}_{\perp!}$ be mixed-variant and locally continuous. An admissible action on n-ary relations for F is a function F^{rel} assigning to each pair $\langle D, E\rangle$ a map

$$F^{rel}_{\langle D,E\rangle} : \mathcal{R}(D) \times \mathcal{R}(E) \to \mathcal{R}(F(D, E)).$$

These maps are compatible with strict morphisms as follows: if $f : D_2 \to D_1$, $g : E_1 \to E_2$, $f^{rel}(R_2) \subseteq R_1$, and $g^{rel}(S_1) \subseteq S_2$, then

$$F(f, g)^{rel}(F^{rel}_{\langle D_1,E_1\rangle}(R_1, S_1)) \subseteq F^{rel}_{\langle D_2,E_2\rangle}(R_2, S_2).$$

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Proposition 5.4.4. The maps $F^{rel}_{\langle D,E\rangle}$ are antitone in the first variable and monotone in the second variable.

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Theorem 5.4.5. Let $\mathbf{D}_{\perp!}$ be a category of domains, and let $F : \mathbf{D}_{\perp!}^{op} \times \mathbf{D}_{\perp!} \to \mathbf{D}_{\perp!}$ be a mixed-variant locally continuous functor with an admissible action on relations. Write $D = \mathrm{FIX}(F)$. If $R, S \in \mathcal{R}^n(D)$ satisfy

$$\mathrm{unfold}^{rel}(R) \subseteq F^{rel}(S, R), \qquad \mathrm{fold}^{rel}(F^{rel}(R, S)) \subseteq S,$$

then $R \subseteq S$. *Proof sketch.* By invariance, id_D is the least fixed point of $\phi(g) = \mathrm{fold} \circ F(g, g) \circ \mathrm{unfold}$. Let

$$P = \{f \in [D \to D] \mid f^{rel}(R) \subseteq S\}.$$

This is admissible. If $g \in P$, then

$$\begin{aligned}\phi(g)^{rel}(R) &= \mathrm{fold}^{rel} \circ F(g, g)^{rel} \circ \mathrm{unfold}^{rel}(R)\\ &\subseteq \mathrm{fold}^{rel} \circ F(g, g)^{rel}(F^{rel}(S, R))\\ &\subseteq \mathrm{fold}^{rel}(F^{rel}(R, S)) \subseteq S.\end{aligned}$$

Thus $\phi(g) \in P$, so $\mathrm{id}_D \in P$, which is exactly $R \subseteq S$.

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Definition 5.4.6. Let F be a mixed-variant functor as above. An admissible action on n-ary relations is called logical if, for all objects D, E and all $R \in \mathcal{R}^n(D)$,

$$F^{rel}_{\langle D,E\rangle}(R, E^n) = F(D, E)^n.$$

Specializing Theorem 5.4.5 to $R = D$ gives the induction principle below.

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Theorem 5.4.7. *(Induction)* Let $\mathbf{D}_{\perp!}$ be a category of domains, and let $F : \mathbf{D}_{\perp!}^{op} \times \mathbf{D}_{\perp!} \to \mathbf{D}_{\perp!}$ be mixed-variant and locally continuous with a logical action on admissible predicates. Let $D = \mathrm{FIX}(F)$. If $S \in \mathcal{R}^1(D)$ is an admissible predicate such that

$$x \in F^{rel}(D, S) \quad \Longrightarrow \quad \mathrm{fold}(x) \in S,$$

then $S = D$. For the primitive constructors, logical actions are given by

$$\begin{aligned} R_\perp &= \mathrm{up}(R) \cup \{\perp\} \subseteq D_\perp, \\ R \times S &= \{\langle x, y\rangle \in D \times E \mid x \in R,\ y \in S\}, \\ [R \to S] &= \{f \in [D \to E] \mid f(R) \subseteq S\}, \\ R \oplus S &= \mathrm{inl}(R) \cup \mathrm{inr}(S) \subseteq D \oplus E. \end{aligned}$$

The actions for composite functors are obtained by composing the corresponding relation actions.

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Definition 5.4.8. Let F be a mixed-variant functor. An admissible action on binary relations is extensional if, for all objects D, E and all $R \in \mathcal{R}^2(D)$,

$$F^{rel}_{\langle D,E\rangle}(R, \sqsubseteq_E) = \sqsubseteq_{F(D,E)} .$$

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Theorem 5.4.9. *(Co-induction)* Let $\mathbf{D}_{\perp!}$ be a category of domains, and let $F : \mathbf{D}_{\perp!}^{op} \times \mathbf{D}_{\perp!} \to \mathbf{D}_{\perp!}$ be mixed-variant and locally continuous with an extensional action on binary relations. Let $D = \mathrm{FIX}(F)$. If $R \in \mathcal{R}^2(D)$ is admissible and, for all $\langle x, y\rangle \in R$,

$$\langle \mathrm{unfold}(x), \mathrm{unfold}(y)\rangle \in F^{rel}(\sqsubseteq_D, R),$$

then $R \subseteq \sqsubseteq_D$.

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Corollary 5.4.10. Two elements of the canonical fixpoint of a mixed-variant locally continuous functor are in the order relation if and only if they are related by a simulation.